One central switch. Existing backbone.

Equipment and logical-network architecture only - no new copper or fibre installation proposed

CENTRAL POINT

LOCAL SERVEURS - ETAGE
One managed central switch aggregates every technical-local switch through the existing backbone and connects the NVR/VMS servers.

TECHNICAL LOCALS

One managed PoE switch at each established technical location, each protected by its own UPS rated at 1000 VA / 600 W minimum.

CONTROL ROOM

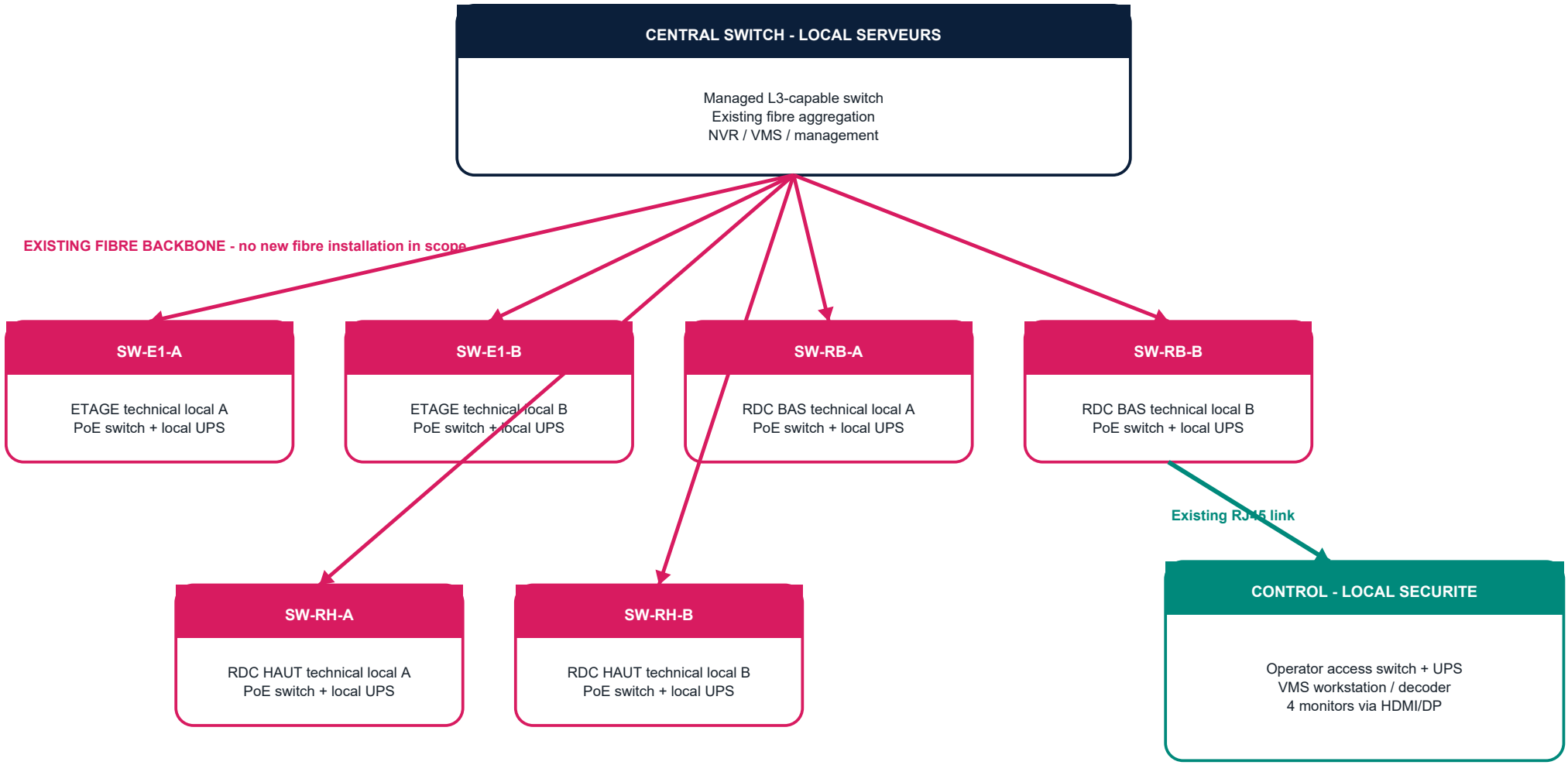
LOCAL SECURITE - RDC BAS
One local operator switch, VMS workstation or decoder, and four displays. Existing RJ45 infrastructure is reused.

CONFIRMED DESIGN BASIS

All copper and fibre links from the technical locations to LOCAL SERVEURS are already installed. The architecture therefore starts at the patch panels and active equipment: local cameras connect to the nearest PoE switch; every local switch reaches the central switch through the existing fibre; the central switch joins the full CCTV system; the NVR/VMS and control-room workstation access the same managed network. Existing links must be tested and labelled, but no new backbone route is included in this design.

Architecture decision: The server-room switch is the only central aggregation point. The switch in the control room is an access/operator switch, not a second core.

Logical architecture



TRAFFIC FLOW

Camera -> nearest local PoE switch -> existing fibre -> central switch -> NVR/VMS. Operators view and manage the system from LOCAL SECURITE. All switching remains under one management and VLAN policy.

FOUR MONITORS

Use one VMS workstation with four GPU outputs or a compatible IP decoder. The screens are display endpoints, not camera switch ports.

Equipment schedule

Location	Equipment	Minimum requirement	Connection
LOCAL SERVEURS - ETAGE	Central switch	Managed L3-capable switch; sufficient SFP/SFP+ ports for every technical-local uplink; VLAN, ACL, STP/RSTP, SNMP; capacity for NVR/VMS traffic	All existing fibre uplinks terminate here
Each technical local	Local PoE switch	Managed PoE+; port count and PoE budget based on cameras in that zone; SFP/SFP+ compatible with installed fibre	Existing fibre to central switch; camera RJ45 locally
Each technical local	Local UPS	1000 VA / 600 W minimum; monitored model preferred; design load <=480 W unless manufacturer permits otherwise	PoE switch and local active accessories
LOCAL SECURITE - RDC BAS	Operator switch	Managed access switch; ports for VMS workstation, decoder and management equipment; local UPS	Existing RJ45 from nearest technical local
LOCAL SECURITE - RDC BAS	Operator station	VMS workstation or decoder with four simultaneous display outputs	Four monitors via HDMI/DisplayPort
LOCAL SERVEURS - ETAGE	NVR/VMS	Storage and throughput sized after camera count, resolution, FPS and retention are confirmed	Central switch server ports

UPS RULE

A 600 W UPS should normally be designed around a maximum continuous load of approximately 480 W to retain headroom. If the PoE switch and cameras exceed that load or require longer autonomy, move to 1500 VA or higher.

POE RULE

For every local switch: required PoE budget >= sum of camera maximum watts x 1.20. Keep spare ports and spare PoE budget for future cameras. Switch rating and UPS rating are separate checks.

EXISTING LINKS

Before connection, identify both ends, clean fibre connectors, verify optical levels, test every copper link, and confirm that installed SFP optics match the fibre type and link speed.

Network configuration and acceptance

Logical segment	Example VLAN	Devices	Policy
CCTV cameras	120	All IP cameras	Reach NVR/VMS, DNS/NTP and approved management only; no general Internet access
VMS / operators	121	NVR/VMS, control-room workstation and decoder	Reach cameras and recording services; role-based operator access
Infrastructure management	199	Central/local switches and UPS network cards	Accessible only from authorised administration hosts; SNMPv3, syslog and NTP

CENTRAL SWITCH

Create the VLAN gateways and ACLs at the central switch or upstream firewall. Mark every existing technical-local uplink as a controlled trunk. Enable loop protection; do not connect a second uncontrolled path from the control room.

DEVICE MANAGEMENT

Use unique credentials, HTTPS/SSH, current firmware, fixed IPs or DHCP reservations, central NTP, configuration backups and a maintained IP/port/camera register.

RECORDING

Confirm camera count, codec, resolution, frame rate, bitrate and retention before selecting storage. Test live view, playback, export, alarm display and time synchronisation from the control room.

Acceptance checklist	Required result
Existing fibre and RJ45 identification	Both ends labelled and mapped to the correct switch ports
Link tests	Copper certification and fibre optical-loss results recorded
UPS test	Actual load and runtime recorded for every technical local
Network test	VLANs, ACLs, management access and loop protection verified
CCTV test	Every camera records, plays back and exports correctly from LOCAL SECURITE

Scope: Active equipment and logical configuration only. Existing backbone cabling is reused; testing and labelling are acceptance activities, not new installation works.